

of the MCU should go low and the standby LED should turn off, and vice versa, on pressing the same key again. If this doesn't happen, replace the microcontroller.

Connect LED1 through LED6 to connector CON1. Press 'Power' key on the remote to switch on the switcher. Page-3 text should appear on the LCD, showing camera-1 is 'on,' the switcher is in manual mode and the default time of camera hold for auto mode. At the same time, Port-C pin RC7 should go high and LED1 should glow. Select other cameras either by pressing keys 1 through 6 or camera up/down keys of the remote and check whether the corresponding LEDs (LED1 through LED6) are glowing (refer to Port-C truth table). Simultaneously, page-3 should show the corresponding camera number

accordingly. If the LEDs don't glow according to the truth table, replace the microcontroller.

When camera-1 is selected, the LCD should show camera-1 and LED1 should glow. If the monitor does not show the video of camera-1, or you hear no audio from the power amplifier, check 5V logic at pins 5 and 6 of IC4. If there is no voltage at these pins, check the availability of 0.6V at the base of transistor T5, +5V at its emitter and +12V at its collector. Follow the same procedure for all other cameras and their corresponding switching circuits.

Software

The source program is written in Assembly language and compiled using MPASM tool suite. The explanation of the Assembly source code is included

in this month's EFY-CD.

Construction

A single-side, solder-side PCB layout for the microcontroller-based remote-controlled six-camera CCTV switcher is shown in Fig. 8 and its component layout in Fig. 9.

Connect CCTV cameras in the sockets provided (SK1 through SK6). Also connect the LCD and remaining components (as shown in Fig. 2) and +5V and +12V from the power supply circuit. Now a welcome message will appear. Press 'Power' key on the remote to start the switcher.

EFY note. The complete project folder containing the source code and other related files is included in this month's EFY-CD and is also available on www.efymag.com. ●

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